



Original research

The influence of baseball pitching distance on pitching biomechanics, pitch velocity, and ball movement



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ABSTRACT

Objectives: To determine whether increasing pitching distance for adult baseball pitchers would affect their upper extremity kinetics, full-body kinematics, and pitched ball kinematics (ball velocity, duration of ball flight, vertical and horizontal break, strike percentage).

Design: Controlled laboratory study.

Methods: Twenty-six collegiate baseball pitchers threw sets of five full-effort fastballs from three different pitching distances (18.44 m, 19.05 m, 19.41 m) in a randomized order. Ball velocity, horizontal and vertical break, duration of ball flight, and strike percentage were computed by a ball tracking system, while pitching kinetics and kinematics were calculated with a 12-camera optical motion capture system. Repeated measures analysis of variance was utilized to detect significant differences among the three different pitching distances ($p < 0.05$).

Results: No significant differences in pitching kinetics and kinematics were observed among the varying pitching distances. Ball velocity and strike percentage were also not significantly different among the pitching distances, however, the duration of ball flight and horizontal and vertical break significantly increased with pitching distance.

Conclusions: Increasing pitching distance may not alter upper extremity kinetics, full-body kinematics, ball velocity or strike percentage in adult pitchers. However, as pitching distance increases the duration of ball flight and amount of horizontal and vertical break also increase. Increased ball flight duration could be an advantage for the hitter while increased ball break could help the pitcher. In conclusion, it is unlikely that moving the mound backwards would significantly affect pitching biomechanics and injury risk; however, the effects on pitching and hitting performance are unknown.

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Practical implications

- Increasing pitching distance does not significantly increase upper extremity pitching kinetics.
- Increasing pitching distance does not significantly alter pitching kinematics.
- Increasing pitching distance increases the duration of ball flight, as well as magnitudes of horizontal and vertical break.
- Increased flight time may help batters but the increased ball break may help pitchers, therefore the overall effect of pitching distance on offensive performance cannot be fully determined from the current study.

- Implementation of increased pitching distance in pilot leagues would add valuable information.

1. Introduction

Lowering the mound is commonly suggested approach to increase offense in baseball^{1,2}; however, results of a recent study raise doubts as to whether such a change would have the desired effect.³ A more recent suggestion has been to move the mound backwards, farther from home plate.¹ The theory is that increased pitch velocity in today's game has reduced the time for the ball to reach home plate, diminishing the ability for the batter to react and hit the ball. Thus, increasing the distance between the pitching mound and home plate may provide the batter with more time to see the ball, swing, and make good contact.

While increasing the pitching distance may increase offense, the effect on pitching mechanics, joint kinetics, and injury risk must be

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considered. This concern is supported by a recent study of adolescent baseball pitchers that reported increased shoulder force and torque when pitching from greater distances, despite no significant change in ball velocity.⁵

Therefore, the primary purpose of this study was to determine the effect of pitching distance on adult baseball pitching biomechanics. Based upon the previous study of adolescent pitchers, it was hypothesized that increased pitching distance would result in increased upper extremity kinetics and perhaps some changes in full-body kinematics. The secondary purpose was to determine whether increasing pitching distance would affect the kinematics of the ball flight (ball velocity, duration of ball flight, horizontal and vertical break, strike percentage). Based upon the previous study of adolescent pitchers, it was hypothesized that ball velocity would not be significantly affected by pitching distance. However, it was hypothesized that as pitching distance increased, duration of ball flight, horizontal break, and vertical break would increase, while the percentage of strikes thrown would decrease.

2. Methods

This study was approved by Sterling Institutional Review Board. A total of 26 healthy, active collegiate pitchers (10 Division 1, 2 Division 2, 11 Division 3, 3 Junior College) were recruited to participate in this study. The pitchers were (mean $\pm$ SD) 21 ± 1 years old, 184 ± 5 cm tall, and had a mass of 90 ± 9 kg. Of the pitchers tested, 21 were right-handed and five were left-handed. Testing was conducted in an indoor biomechanics laboratory at the American Sports Medicine Institute (ASMI).

After informed consent was obtained, each participant was instructed to change into tight-fitting spandex shorts, socks and turf shoes, and a baseball hat. Anthropometric measures were collected and retro-reflective markers were placed on the participant's bony landmarks as previously described^{5–7} with one additional marker on the heel of the foot on the pitcher's throwing arm side. Once the markers were attached, the participant was instructed to warm-up as he normally would for a game. Each participant's warm-up included stretching and non-throwing drills, followed by a self-selected number of warm-up pitches from the mound. After conclusion of the warm-up, each participant threw sets of eight full-effort fastballs from three different pitching distances (18.44 m, 19.05 m, 19.41 m) in a randomized order. Each pitch was thrown from a pitching mound to a catcher positioned behind home plate located at the appropriate distance. The shortest distance of 18.44 m (60.5 ft) is the current official distance for MLB and other adult baseball leagues. The other two distances were suggested to the investigators by MLB officials. The suggested distances correspond with moving the mound backwards 0.61 m (2.0 ft) and 0.97 m (3.2 ft) from the current pitching distance. When the pitching rubber is at the farthest distance (19.41 m), it is located at the geometric center of the infield diamond, halfway between home plate and second base.

Three-dimensional motion data of each pitcher were tracked with a 12-camera automated motion capture system sampling at 240 Hz with a three-dimensional residual threshold set at 0.45 ± 0.20 mm (Raptor System, Motion Analysis Corporation, USA). In addition, ball kinematics were measured for the final 21 participants using a PITCHf/x system (SportsMEDIA Technology, Fremont, CA, USA). For each pitcher tested, the first three pitches of each condition were excluded from the analysis to allow the pitcher to fully adapt to the pitching distance. The participant was unaware that these pitches would be excluded from analysis. Thus, pitcher biomechanics and ball kinematics were calculated for the final five pitches for each condition.

Thirty-seven pitcher kinematic and kinetic variables were calculated as previously described.^{7–9} Initial ball velocity was measured at a location 3.05 m (10 feet) in front of the pitching rubber. Final ball velocity was measured as the ball crossed the front of home plate. The duration of ball flight was defined as the total time it took the baseball to travel between these two locations. Horizontal break was defined as the maximum horizontal deviation from an imaginary straight-line path between the release point and the point where the pitch crossed the back tip of home plate. Vertical break was defined as the maximum vertical deviation from an imaginary straight-line path between the release point and the point where the pitch crossed the back tip of home plate. For each pitching distance, strike percentage was calculated as the number of pitches within the PITCHf/x strike zone divided by the total number of pitches (i.e., both in and out of the zone). For each of the three pitching distance conditions, the magnitudes of all investigated parameters were averaged for each participant. Repeated measures analysis of variance (ANOVA) was utilized to detect significant differences among the pitching distances. When a significant difference ($p < 0.05$) was found, pairwise differences were analyzed with a Tukey post-hoc test ($p < 0.05$).

3. Results

No significant pitcher kinetic (Tables 1) or kinematic (Table 2) differences were observed among the three distances. Initial ball velocity, final ball velocity, and strike percentage showed no significant differences among the pitching distances, but duration of ball flight, horizontal break, and vertical break significantly increased as pitching distance increased (Table 3).

4. Discussion

The hypothesis that joint kinetics would increase with pitching distance was not supported, as there were no significant differences found. This contrasts with a previous study of adolescent pitchers that found significant increases in shoulder anterior force and horizontal adduction torque when pitching from farther distances.⁵ A probable explanation for the apparent contradiction of findings is that the changes in pitching distances were much more extreme in the adolescent study. The adolescent study increased pitching distance from 14.02 m (standard youth distance) to 16.46 m and 18.44 m, whereas the current study varied the distance from 18.44 m (standard adult distance) to 19.05 m and 19.41 m. That corresponds to a 32% (4.42 m) increase in distance for the adolescent pitchers, but only a 5% (0.97 m) increase for the adults. It was also discussed in the previous study that youth pitchers may have been throwing with increased effort and compensation techniques (e.g. decreased trunk forward trunk tilt at ball release) at farther distances in order to ensure the baseball reached home plate. With the current study examining the effect of pitching distance in adult (collegiate) baseball pitchers, these compensation techniques may not be utilized due to advancements in skeletal maturity and muscular strength. The finding that shoulder and elbow kinetics did not increase for adult pitchers at the farther distances suggests that moderate changes in pitching distance may not significantly increase shoulder and elbow stress.

The hypothesis that pitching kinematics would be significantly different with increased pitching distance was not supported in this study. The findings for the current study are not surprising, as the previous study examining the effect of pitching distance in adolescent baseball pitchers found only minimal kinematic changes.⁵ Therefore, if MLB decided to move the mound back slightly, results from this study suggest pitchers would not adjust their fastball mechanics.

Table 1

Maximum shoulder and elbow kinetic values from varying distances (m ± sd).

	18.44 m	19.05 m	19.41 m	Standard error	P value
Shoulder kinetics					
Shoulder anterior force (N)	368 ± 65	368 ± 65	369 ± 66	12.8	0.75
Shoulder proximal force (N)	1047 ± 112	1043 ± 117	1045 ± 116	22.6	0.65
Shoulder horizontal adduction torque (N*m)	104 ± 21	105 ± 21	104 ± 20	4.2	0.63
Shoulder internal rotation torque (N*m)	85 ± 17	85 ± 17	85 ± 17	3.4	0.73
Elbow kinetics					
Elbow proximal force (N)	1007 ± 108	1007 ± 108	1007 ± 111	22.4	0.97
Elbow varus torque (N*m)	84 ± 17	84 ± 18	84 ± 18	3.5	0.71
Elbow flexion torque (N*m)	60 ± 9	59 ± 9	60 ± 9	1.8	0.53

Table 2

Pitcher kinematics from varying pitching distances (m ± sd).

	18.44 m	19.05 m	19.41 m	Standard error	P value
Foot contact					
Stride length (% height)	82 ± 3	82 ± 3	82 ± 3	0.7	0.38
Lead foot position (cm)	19 ± 12	19 ± 11	19 ± 12	2.4	0.48
Lead foot angle (°)	6 ± 10	6 ± 10	5 ± 10	2.0	0.46
Lead knee flexion (°)	44 ± 8	44 ± 8	44 ± 8	1.6	0.83
Pelvis rotation (°)	33 ± 13	33 ± 12	33 ± 13	2.6	0.48
Trunk separation (°)	46 ± 11	46 ± 11	46 ± 11	2.3	0.83
Upper trunk tilt (°)	9 ± 6	9 ± 6	10 ± 6	1.3	0.20
Shoulder abduction (°)	92 ± 9	92 ± 10	92 ± 9	2.0	0.73
Shoulder horizontal abduction (°)	16 ± 11	16 ± 12	17 ± 11	2.3	0.85
Shoulder external rotation (°)	62 ± 31	63 ± 30	62 ± 31	6.1	0.67
Elbow flexion (°)	97 ± 19	98 ± 19	97 ± 19	3.9	0.29
Arm cocking phase					
Maximum shoulder external rotation (°)	162 ± 10	162 ± 10	162 ± 10	2.1	0.59
Maximum shoulder horizontal adduction (°)	20 ± 6	20 ± 6	20 ± 5	2.0	0.84
Maximum elbow flexion (°)	108 ± 13	108 ± 14	107 ± 12	2.7	0.06
Ball release					
Lead knee flexion (°)	39 ± 14	39 ± 14	37 ± 14	2.8	0.88
Trunk forward tilt (°)	31 ± 6	32 ± 6	31 ± 6	1.3	0.25
Trunk side tilt (°)	22 ± 9	22 ± 9	23 ± 8	1.8	0.61
Shoulder abduction (°)	88 ± 9	88 ± 9	88 ± 9	1.9	0.45
Elbow flexion (°)	22 ± 3	23 ± 4	22 ± 4	0.8	0.36
Arm slot (°)	55 ± 7	54 ± 7	54 ± 7	1.5	0.14
Maximum angular velocities					
Pelvis rotation velocity (°/s)	572 ± 81	571 ± 81	570 ± 79	15.8	0.82
Upper trunk rotation velocity (°/s)	1129 ± 96	1125 ± 97	1124 ± 96	19.0	0.12
Knee extension velocity (°/s)	289 ± 111	287 ± 124	281 ± 106	22.4	0.60
Elbow extension velocity (°/s)	2696 ± 273	2706 ± 287	2691 ± 295	55.9	0.40
Shoulder internal rotation velocity (°/s)	6816 ± 787	6800 ± 785	6801 ± 795	154.8	0.83

Table 3

Ball velocity and ball flight duration (m ± sd).

	18.44 m	19.05 m	19.41 m	Standard error	P value
Ball kinematics					
Initial ball velocity (m/s)	37 ± 1	37 ± 1	37 ± 1	0.70	0.36
Final ball velocity (m/s)	33 ± 1	33 ± 1	33 ± 1	0.61	0.25
Ball flight duration (s) ^{a,b,c}	0.43 ± 0.02	0.45 ± 0.02	0.46 ± 0.02	0.004	<0.001
Horizontal break (cm) ^a	2.69 ± 7.1	3.30 ± 7.6	3.91 ± 7.9	0.64	<0.001
Vertical break (cm) ^{a,b}	15.0 ± 2.8	16.6 ± 3.3	16.8 ± 3.3	0.27	<0.001
Strike percentage (%)	47 ± 26	32 ± 20	34 ± 21	0.05	0.09

^a Significant post-hoc difference between 18.44 m and 19.05 m distance.^b Significant post-hoc difference between 18.44 m and 19.41 m distance.^c Significant post-hoc difference between 19.05 m and 19.41 m distance.

The hypotheses that ball velocity would not be significantly different and that the duration of ball flight and horizontal and vertical break would significantly increase, were all supported. When comparing the duration of ball flight between the standard pitching distance (18.44 m) and the increased pitching distances (19.05 m & 19.41 m), the duration of ball flight increased 0.02 and 0.03 s, respectively. In addition to the change in duration of ball flight, the magnitudes of horizontal and vertical break significantly increased as well. For horizontal break, the post-hoc analysis indicated a

significant difference when comparing the 18.44 m and 19.41 m pitching distance, resulting in 1.3 cm more horizontal break when pitching from the furthest distance. When comparing the magnitudes of vertical break between the standard pitching distance and the increased pitching distances, the magnitude of vertical break increased by 1.6 and 1.8 cm, respectively. Thus, increasing the pitching distance may allow the batter more time to react to the pitch, however the horizontal and vertical break of the pitch may increase as well, counteracting the timing effect.

While the results of the current study suggest moving the mound back would allow a batter more time to see and react to the pitch without increasing the stress on the throwing arm, there are limitations to this study. First, only fastballs were analyzed. While previous research has shown similar joint kinetics between fastballs and breaking pitches,^{6,10} it is widely known that ball kinematics are different. Further research should examine the changes in ball movement for other pitch types from varying distances. Next we found that while duration of ball flight for fastballs increased with pitching distance, the magnitude of break also increased as well. Therefore, increased pitch distance benefits batters with extra swing time but benefits pitchers by allowing more pitch movement. A change in pitching distance may also affect other aspects of the game, such as the number of pitches required per game and the amount of fielding required from pitchers on bunts, ground balls, and covering first base. While changes in pitching biomechanics and ball velocity were not detected in this laboratory study, it is unknown whether pitchers would alter their mechanics over time when adapting to a longer pitching distance. It would be valuable to supplement the laboratory data presented here with pilot field data from an adult baseball league(s) using increased pitching distance.

5. Conclusion

The current study suggests that a moderate (up to 1 m) increase in pitching distance has no statistically significant effect on pitching kinematics or kinetics. The finding that the duration of ball flight increased with pitching distance supports the concept that moving the mound back would give batters more time to decide and execute their swing. However, the magnitudes of horizontal and vertical break also increased, possibly counteracting the effect of increasing the distance to improve offensive performance. Therefore, while it is unlikely that moving the mound backwards would significantly affect pitching biomechanics and injury risk, the potential effect on offensive performance cannot be fully determined from the current study.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jsams.2020.02.005>.

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